

Product rule, quotient rule, and trig functions

Question: Suppose $f(x)$ and $g(x)$ are differentiable.

What is $\frac{d}{dx}(f \cdot g)$?

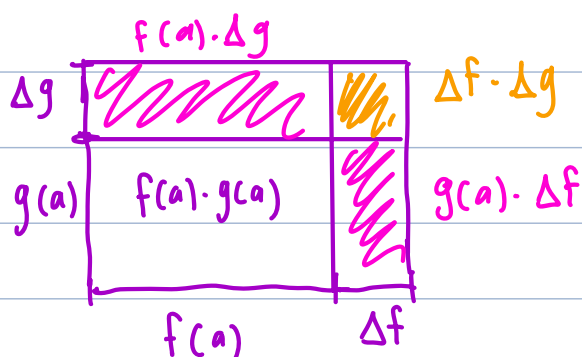
Idea: Suppose we change x from $x=a$ to $x=a+\Delta x$ small change in x

How does $f \cdot g$ change in response?

f : $f(a+\Delta x) = f(a) + \Delta f$ "change in f , in response to change in x "

g : $g(a+\Delta x) = g(a) + \Delta g$

$f \cdot g$: $(f \cdot g)(a+\Delta x) = (f(a) + \Delta f)(g(a) + \Delta g)$



$$= f(a) \cdot g(a) + \Delta f \cdot g(a) + f(a) \cdot \Delta g + \Delta f \cdot \Delta g$$

$$\Delta(f \cdot g) = f \cdot g(a+\Delta x) - f \cdot g(a)$$

$$= \left(\cancel{f(a)g(a)} + \Delta f \cdot g(a) + f(a) \cdot \Delta g + \Delta f \cdot \Delta g \right) - \cancel{f(a)g(a)}$$

$$\frac{d}{dx}(f \cdot g) = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta(f \cdot g)}{\Delta x} \right)$$
$$= \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta f \cdot g(a)}{\Delta x} + f(a) \cdot \frac{\Delta g}{\Delta x} + \frac{\Delta f \cdot \Delta g}{\Delta x} \right)$$

Annotations: $\frac{\Delta f \cdot g(a)}{\Delta x} \rightarrow f'(a)$, $f(a) \cdot \frac{\Delta g}{\Delta x} \rightarrow g'(a)$, $\frac{\Delta f \cdot \Delta g}{\Delta x} \rightarrow 0$

Product rule:

$$\frac{d}{dx}(f \cdot g)(a) = f'(a) \cdot g(a) + f(a) \cdot g'(a)$$

$$\frac{d}{dx}(f \cdot g) = f' \cdot g + f \cdot g'$$

Example: Calculate $\frac{d}{dx}(x^2 e^x)$

Answer:

$$\frac{d}{dx}(x^2 \cdot e^x) = \underbrace{\frac{d}{dx}(x^2)}_{2x} \cdot e^x + x^2 \cdot \underbrace{\frac{d}{dx}(e^x)}_{e^x}$$

↑
Product rule

$$= 2x \cdot e^x + x^2 \cdot e^x$$

$$= \boxed{(2x + x^2) e^x}$$

Quotient rule: Suppose f and g are differentiable.

$$\text{Then } \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f' \cdot g - f \cdot g'}{g^2}$$

Example: Find $\frac{d}{dx} \left(\frac{x-3}{x^2+1} \right)$

$$f = x-3 \Rightarrow f' = \frac{d}{dx}(x-3) = \underbrace{\frac{d}{dx}(x)}_1 - \underbrace{\frac{d}{dx}(3)}_0 = 1$$

$$g = x^2+1 \Rightarrow g' = \frac{d}{dx}(x^2+1) = \underbrace{\frac{d}{dx}(x^2)}_{2x} + \underbrace{\frac{d}{dx}(1)}_0 = 2x$$

$$\frac{d}{dx} \left(\frac{x-3}{x^2+1} \right) = \frac{f' \cdot g - f \cdot g'}{g^2}$$

$$= \frac{1 \cdot (x^2+1) - (x-3) \cdot 2x}{(x^2+1)^2} = \frac{x^2+1 - 2x^2+6x}{(x^2+1)^2}$$

$$= \boxed{\frac{-x^2+6x+1}{(x^2+1)^2}}$$

Product of three functions

$$\frac{d}{dx} (f \cdot \underbrace{g \cdot h}_{\text{product rule}}) =$$

$$f' \cdot (g \cdot h) + f \cdot \underbrace{(g \cdot h)'}_{g'h + gh' \text{ (product rule)}}$$

$$= f'gh + f(g'h + gh')$$

$$= \boxed{f'gh + fg'h + fgh'}$$

$$\frac{d}{dx} (f \cdot g) = f' \cdot g + f \cdot g'$$

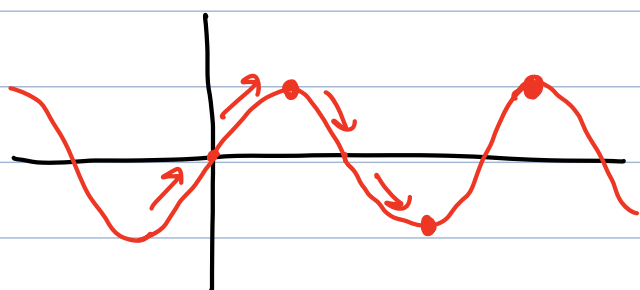
treat $(g \cdot h)$ as g

Four Functions:

$$(f \cdot g \cdot h \cdot k)'$$

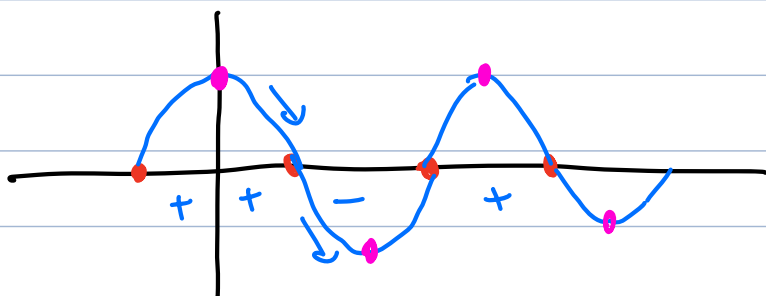
$$= f'ghk + fg'hk + fgh'k + fghk'$$

Derivatives of sin and cos



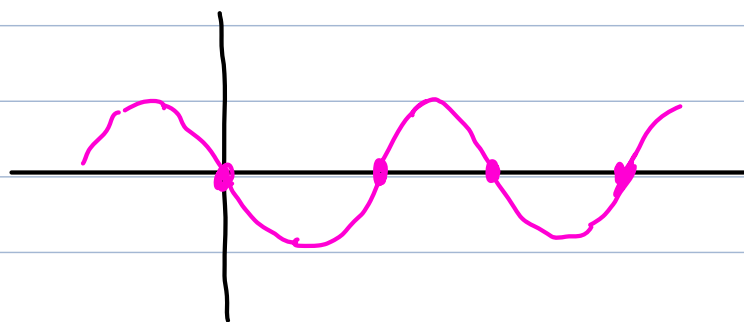
$$y = \sin x$$

derivative



$$y = \cos x$$

derivative



$$y = -\sin x$$

in fact,

$$\boxed{\begin{aligned} \frac{d}{dx} (\sin x) &= \cos(x) \\ \frac{d}{dx} (\cos x) &= -\sin(x) \end{aligned}}$$

Example

what is the seventh derivative of cos?

$$\cos^{(7)}(x) = ?$$

$$\begin{aligned} \cos(x) &\xrightarrow{\text{deriv.}} -\sin(x) \xrightarrow{\text{deriv.}} -\cos(x) \xrightarrow{\text{deriv.}} -(-\sin(x)) = \sin(x) \\ &\xrightarrow{\text{deriv.}} \cos(x) \xrightarrow{\text{deriv.}} -\sin(x) \xrightarrow{\text{deriv.}} -\cos(x) \xrightarrow{\text{deriv.}} \boxed{\sin(x)} \end{aligned}$$

Example a) What is $\frac{d}{dx}(x \sin(x))$?

b) $\frac{d}{dx}(\sqrt{x} \cdot \cos(x) \cdot e^x)$?

$$\begin{aligned} \text{a) } \frac{d}{dx}(x \sin(x)) &= \underbrace{\frac{d}{dx}(x)}_1 \cdot \sin(x) + x \cdot \underbrace{\frac{d}{dx}(\sin x)}_{\cos x} \\ &= 1 \cdot \sin x + x \cdot \cos x = \boxed{\sin x + x \cos x} \end{aligned}$$

b) $\frac{d}{dx}(f \cdot g \cdot h) = f'gh + fg'h + fgh'$

$$\frac{d}{dx} \left(\underbrace{\sqrt{x}}_f \cdot \underbrace{\cos(x)}_g \cdot \underbrace{e^x}_h \right) =$$

$$\begin{aligned} &\underbrace{\frac{d}{dx}(\sqrt{x})}_{\frac{d}{dx}(x^{1/2})} \cdot \cos(x) \cdot e^x + \sqrt{x} \cdot \underbrace{\frac{d}{dx}(\cos x)}_{-\sin x} \cdot e^x + \sqrt{x} \cdot \cos x \cdot \underbrace{\frac{d}{dx}(e^x)}_{e^x} \\ &= \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}} \end{aligned}$$

$$= \boxed{\frac{\cos(x) \cdot e^x - \sqrt{x} \sin x e^x + \sqrt{x} \cos x e^x}{2\sqrt{x}}}$$

Trig limits

Remember

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

This is connected to derivatives of trig functions

$$\frac{d}{dx}(\sin(x)) \bigg|_0 = \lim_{h \rightarrow 0} \frac{\sin(\overbrace{0+h}^h) - \underbrace{\sin(0)}_0}{h}$$

$$\cos(x) \bigg|_0$$

$$\cos(0) = 1$$

$$= \lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1$$

Example

Find

$$\lim_{x \rightarrow 0} \frac{\sin(\overbrace{4x}^u)}{7x}$$

Substitution

$$u = 4x$$

$$7x = \frac{\overbrace{4x}^u}{4} \cdot 7 = \frac{7u}{4}$$

As $x \rightarrow 0$, $u = 4x \rightarrow 0$

$$\lim_{x \rightarrow 0} \frac{\sin(\overbrace{4x}^u)}{7x} = \lim_{u \rightarrow 0} \frac{\sin(u)}{7 \frac{u}{4}} = \lim_{u \rightarrow 0} \frac{4 \sin(u)}{7u}$$

$$= \frac{4}{7} \left(\lim_{u \rightarrow 0} \frac{\sin(u)}{u} \right) = \frac{4}{7} \cdot 1 = \boxed{\frac{4}{7}}$$

Example: find $\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h}$

$$\frac{\cos(0) - 1}{0} = \frac{1 - 1}{0} = \frac{0}{0} \text{ indeterminate}$$

idea: use the identity

$$\cos^2 h + \sin^2 h = 1$$

$$\cos^2 h - 1 = -\sin^2 h$$

$$(\cos(h) - 1)(\cos(h) + 1)$$

$$\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} = \lim_{h \rightarrow 0} \frac{(\cos(h) - 1)(\cos(h) + 1)}{h(\cos(h) + 1)}$$

$$= \lim_{h \rightarrow 0} \frac{\cos^2(h) - 1}{h(\cos(h) + 1)} = \lim_{h \rightarrow 0} \frac{-\sin^2 h}{h(\cos(h) + 1)}$$

$$= \left(\lim_{h \rightarrow 0} \frac{\sin h}{h} \right) \left(\lim_{h \rightarrow 0} \frac{-\sin h}{\cos(h) + 1} \right)$$

1

direct subst. $h=0$

$$\frac{0}{1+1} = \frac{0}{2} = 0$$

$$= 1 \cdot 0 = 0$$

$$\Rightarrow \boxed{\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} = 0}$$

We've seen: $\frac{d}{dx}(\sin(x)) = \cos(x)$

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

Let's now see why this is true.

$$\sin'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(\sin x \cos h + \cos x \sin h) - \sin(x)}{h}$$

$\sin(\alpha + \beta)$
 $= \sin \alpha \cos \beta$
 $+ \cos \alpha \sin \beta$
"angle addition",
formula

$$= \lim_{h \rightarrow 0} \frac{\sin x (\cos(h) - 1) + \cos x \cdot \sin(h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x (\cos(h) - 1)}{h} + \lim_{h \rightarrow 0} \frac{\cos(x) \cdot \sin(h)}{h}$$

$$= \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} + \cos(x) \cdot \lim_{h \rightarrow 0} \frac{\sin(h)}{h}$$

$$= \sin(x) \cdot 0 + \cos(x) \cdot 1 =$$

$$\boxed{\cos(x)}$$

$$\Rightarrow \boxed{\sin'(x) = \cos x}$$

As an exercise, you can try to prove
(similar ideas to $\frac{d}{dx}(\sin(x))$)

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

Derivatives of other trig functions

Example: $\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$

quotient rule: $\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$

$f = \sin x \Rightarrow f' = \cos x$

$g = \cos x \Rightarrow g' = -\sin x$

$$\frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2(x)} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} = \left(\frac{1}{\cos x}\right)^2 = \boxed{\sec^2 x}$$

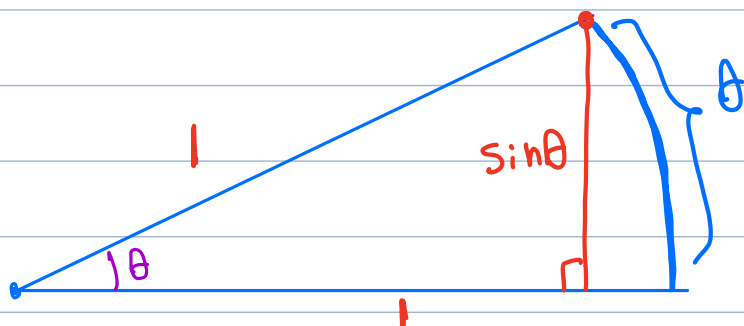
$\underbrace{\frac{1}{\cos x}}_{\sec x}$

Summarizing derivatives of trig functions:

$f(x)$	$f'(x)$
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$\tan(x)$	$\sec^2(x)$
$\sec(x)$	$\sec(x)\tan(x)$
$\cot(x)$	$-\csc^2(x)$
$\csc(x)$	$-\csc(x)\cot(x)$

One missing piece: why $\lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} = 1$?

Let's draw a picture.



consider a sector of a circle with angle θ , radius 1

reminder:

circumference

$$= 2\pi r$$

length of segment of angle θ

$$= \theta r$$

(in our case, $r = 1$)

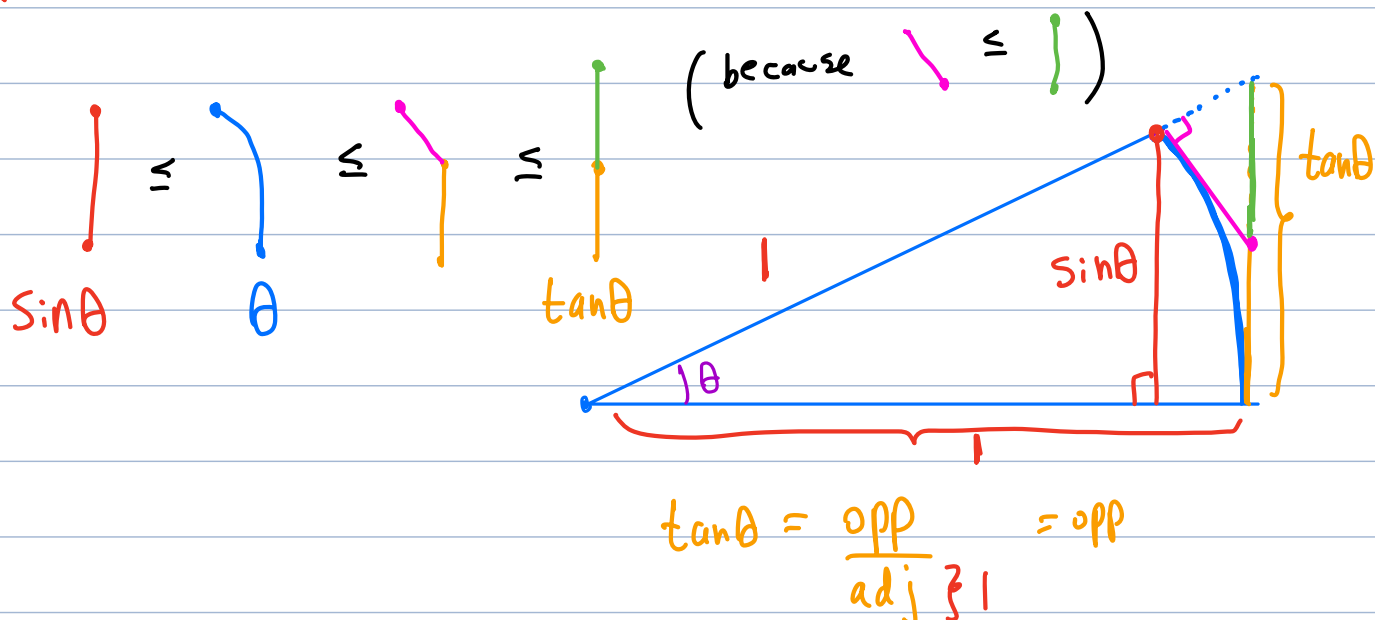
intuition:



as θ gets small,
segment of circle close to vertical

$$\sin \theta \approx \theta \Rightarrow \frac{\sin \theta}{\theta} \approx 1$$

Idea:



$$\sin \theta \leq \theta \leq \tan \theta$$

divide everything by $\sin \theta$

$$1 \leq \frac{\sin \theta}{\sin \theta} \leq \frac{\theta}{\sin \theta} \leq \frac{\tan \theta}{\sin \theta} \quad \frac{\sin \theta / \cos \theta}{\sin \theta} = \frac{1}{\cos \theta}$$

$$1 \leq \frac{\theta}{\sin \theta} \leq \frac{1}{\cos \theta}$$

$\downarrow$ $\downarrow$ as $\theta \rightarrow 0$
 1 1

$\Rightarrow$

$$\lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$$

by squeeze theorem

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

(reciprocal of above)